The Mirror Design Pattern: Strict Data Geometry over Model Scale for Prompt Injection Detection

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Abstract

Prompt injection defenses are often framed as semantic understanding problems and delegated to increasingly large neural detectors. For the first screening layer, however, the requirements are different: the detector runs on every request and therefore must be fast, deterministic, non-promptable, and auditable. We introduce Mirror, a data-curation design pattern that organizes prompt injection corpora into matched positive and negative cells so that a classifier learns control-plane attack mechanics rather than incidental corpus shortcuts. Using 5,000 strictly curated open-source samples—the largest corpus supportable under our public-data validity contract—we define a 32-cell mirror topology, fill 31 of those cells with public data, train a sparse character n-gram linear SVM, compile its weights into a static Rust artifact, and obtain 95.97% recall and 92.07% F1 on a 524-case holdout at sub-millisecond latency with no external model runtime dependencies. On the same holdout, our next line of defense, a 22-million-parameter Prompt Guard 2 model reaches 44.35% recall and 59.14% F1 at 49 ms median and 324 ms p95 latency. Linear models still leave residual semantic ambiguities such as use-versus-mention for later pipeline layers, but within that scope our results show that for L1 prompt injection screening, strict data geometry can matter more than model scale.

1 Introduction

Prompt injection defenses are increasingly built around semantic models that attempt to interpret whether a string of text is malicious. The framing is understandable. Prompt injection is written in natural language, so it is tempting to assume that the detector should also be a language model. The problem is architectural. The first detector in the path runs on every request. It becomes part of the security boundary. At that point the detector needs properties that are not captured by a benchmark score alone: low latency, stable behavior, transparent failure modes, and resistance to prompt-based manipulation.

Ayub and Majumdar's CAMLIS 2024 result [1] raised the key question for this project. Their low-dimensional projections of dense prompt embeddings did not suggest a clean linear boundary, which made us ask whether prompt injection was intrinsically resistant to simple classifiers or whether the difficulty was specific to the dense semantic representations they studied. This paper starts from that question: can a linear SVM distinguish benign requests from prompt injection attempts at L1?

The L1 detector should minimize adaptivity, runtime complexity, and prompt-sensitive behavior relative to the model it is protecting. It should be static, discriminative, and easy to inspect. A promptable model—one whose behavior can be altered by the content it is classifying—in the hot path can expand the attack surface by inserting another instruction-following system into the security boundary. That may be acceptable for later layers that process a small residual, but it is a poor default for a first-pass screen. In our implementation the L1 weights are compiled into the binary as a sparse perfect hash map, which puts the detector in a different operational and attack-surface class from a semantic model.

The main obstacle to this approach was not lack of model capacity. It was lack of geometric discipline in the data. Public prompt injection corpora are mixed in language, length, topic, formatting, and label quality. If malicious and benign examples are not aligned across these nuisance dimensions, a linear classifier learns shortcuts. It memorizes corpus artifacts instead of injection structure. By *geometry*, we mean the intentional, cell-based pairing of contrastive malicious and benign examples across nuisance dimensions such as language, length, topic, and format, so that those variables stop defining the easiest boundary. We call this pattern *Mirror* because each malicious lane is paired with a mirrored benign counterpart under the same cell contract. Mirror was developed as a response to that problem and treats curation as a geometric engineering task.

Our claim is narrow: strict data geometry can move the decision boundary enough that a simple sparse classifier becomes a practical and effective first-line defense for L1 prompt injection screening. The paper supports that claim through three checkpoints: `v2`, which established the initial Mirror baseline; `v3`, which showed that data geometry materially changed performance with the model family held fixed; and `v5`, which imposed a much stricter multilingual validity contract and closed 31 of 32 cells with public data. Accordingly, results are interpreted primarily within each checkpoint rather than as a single directly comparable benchmark table across all versions.

1.1 Task Definition and Scope

The evaluated L1 task is a binary prompt injection screening task, not a general safety or jailbreak benchmark. Positive examples are control-plane attacks: prompts or untrusted payloads that attempt to override instructions, reassign roles, extract hidden system or tool context, smuggle instructions through indirect data, or otherwise alter the governing policy of the protected model. Operationally, Mirror organizes this class into eight reason categories: instruction override, roleplay jailbreak, meta-probe, exfiltration, adversarial suffix, indirect injection, obfuscation, and constraint bypass.

Negative examples are benign requests, including instruction-following tasks and security-adjacent text, that do not attempt to hijack the model's control flow. We explicitly exclude pure content-safety violations that may be unsafe but are not prompt injection—such as toxicity, copyright, PII, and academic dishonesty—from the positive class. Use-versus-mention ambiguity remains unresolved: quoted or discussed attacks are a residual failure mode, as demonstrated in the hard-benign challenge set (§6.7).

Our contributions are:

1. We present Mirror as a reusable design pattern for prompt injection curation, based on matched positive and negative geometric cells.
2. We show that prompt injection can admit a strong linear decision boundary for L1 screening when represented with sparse character n-grams and curated with strict geometry.
3. We introduce a provenance-first curation and evaluation workflow that made leakage, source drift, and false occupancy visible instead of silently inflating results.
4. We show that a 5k Mirror-trained linear SVM can substantially exceed the evaluated transformer baseline on the same held-out set while operating with far lower latency and far simpler deployment requirements.
5. We characterize the residual failure modes—contextual ambiguity, use-versus-mention, and semantically thin attacks—that define the remaining architectural problem for replacing or reducing the L2a semantic layer.

To keep those checkpoints explicit, Table 1 summarizes the versions discussed in the paper and the role each one plays in the argument.

Reading guide. Section 4 focuses on the geometry claim, with `v2` included as the precursor needed to interpret the `v2`→`v3` change under roughly the same model family. Section 5 turns to

Table 1: Checkpoint summary. Each row is a named dataset–model configuration discussed in the paper.

Checkpoint	What changed	Role in argument
`v2` 19k train / 2.4k holdout	Initial Mirror baseline	Geometry baseline (0.835 F1); paired with `v3` to isolate the effect of data quality
`v3` 19k locked / 1.9k holdout	Source locking and provenance cleanup; model family held fixed	Geometry proof point: 0.918→0.926 F1 from data discipline alone
`v5` 5k curated / 524 holdout	Strict multilingual validity contract; 31/32 cell closure	Headline result: 0.921 F1 / 0.960 recall; direct PG2 comparison

`v5`, asking whether the method survives a much stricter validity and multilingual coverage contract. The non-headline checkpoints are therefore included as architectural and methodological landmarks, not as competing final benchmark claims.

2 Related Work

Recent prompt injection detection work has generally assumed that better semantic modeling should yield better detectors. Prompt Guard 2 (Meta, 2024) [2] is the most obvious practical reference point. It is a compact transformer classifier (22M and 86M parameter variants) designed for prompt injection and jailbreak detection, and it serves as the primary baseline in this study because it represents the current "small but semantic" approach. This model is particularly relevant here because it also appears as the next layer of defense (L2a) after the linear classifier (L1) in the Parapet stack. Other public baselines, including ProtectAI's DeBERTa-v3-based prompt injection detector [3] (F1 0.539 on an earlier 2,386-case `v2` holdout), make the same assumption: prompt injection is best handled by a semantic model, even if the model is compressed or deployed as a binary classifier rather than a generative LLM. ProtectAI's training recipe was in fact the starting point for this project's own data sourcing, before the Mirror pattern diverged the approach.

For the direct comparison in this paper, we evaluate the 22M-parameter variant (the smallest and fastest PG2 checkpoint, making it the most favorable baseline for a hot-path screening comparison), recorded in the experiment manifests as `pg2-22m`.

Concurrent work such as PromptScreen [4] reaches a similar architectural conclusion from a different direction. PromptScreen uses a multi-stage pipeline with a TF-IDF linear SVM as the core screening component and reports strong CPU-efficient performance on a corpus of more than 30,000 prompts. That result strengthens the argument that a simple linear detector belongs in the hot path. The difference is that PromptScreen treats the model and pipeline as the main story, while Mirror treats corpus geometry as the main lever and asks why a linear screen can work so well with far less data when the corpus is shaped correctly.

Our results point to a different representation-and-curation regime. We move away from dense semantic embeddings and toward sparse character n-grams, then constrain the corpus geometry so that language, format, and related nuisance variables stop serving as cheap shortcuts. Under that representation and data discipline, the task admits a much cleaner linear boundary for L1 screening, with higher recall than the non-linear models Ayub and Majumdar preferred in their setting.

More broadly, this is not unique to prompt injection. Prior text-classification work has found that TF-IDF linear SVMs can remain competitive with fine-tuned pre-trained language models on domain-specific corpora, suggesting that larger semantic models are not automatically the right default for every classification setting [5].

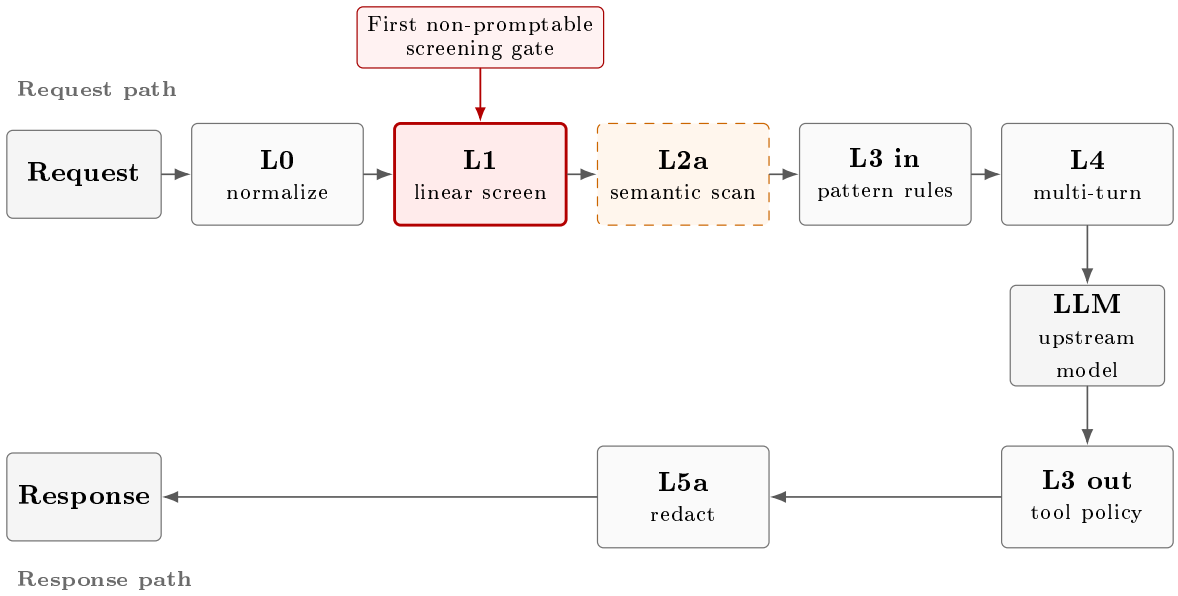

Figure 1: The Parapet layered pipeline. L1 (this paper) applies a compiled sparse linear screen on every request. Later layers handle the residual.

Mirror is adjacent to data-centric work on counterfactually augmented data and spurious-correlation mitigation in NLP [6, 7, 8]. The difference is unit of intervention and goal. Rather than rewriting individual examples into counterfactual variants, Mirror treats the corpus itself as the unit of design. Its aim is not only better classifier behavior, but an auditable training contract: each cell has explicit scope, provenance, and pairing logic.

Finally, this work sits within layered defense architectures rather than replacing them. We do not argue against cascades. In fact, the system design here depends on them. The claim is that the first layer should be deterministic and cheap, while later layers can handle the smaller residual set where semantic context genuinely matters.

3 System Architecture

This establishes the deployment context for the paper's claim: the question is not only whether L1 scores well offline, but whether it can function as a real first-pass gate inside a layered defense.

Parapet began as a layered runtime defense rather than an isolated classifier benchmark. The system established a request-path pipeline: L0 normalizes and assigns trust metadata, L1 applies a lightweight linear screen (the focus of this paper), L2a runs an optional semantic scan for residual ambiguity, L3 enforces deterministic pattern and policy rules, and L4 adds multi-turn context before the request reaches the upstream LLM. A symmetric response path handles output redaction and tool-call policy. The practical goal was not to build a perfect detector in one shot. It was to create a security boundary that could run on every request without turning the detector itself into an operational or attack-surface liability.

The L1 layer was simple from the start and remained simple in the final system. Because the L1 architecture did not change across checkpoints, we describe the `v5` form here. In its mature 5k form, training uses `CountVectorizer` over sparse character word-boundary n-grams with a `LinearSVC`, and the best run used $n = 3..5$, `min_df=5`, binary presence, $C = 1.0$, and no class weighting. At runtime, the learned coefficients are compiled into the Rust binary as a `phf` perfect hash map rather than shipped as an opaque external model. Inference is a single bias-initialized dot product over extracted n-grams, with the raw margin also passed through a

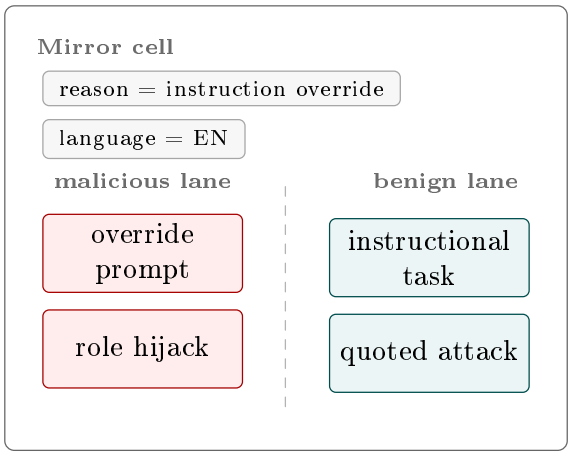

Figure 2: From Mirror cell contract to compiled L1 artifact. Each cell defines matched positive and negative examples; the SVM boundary is compiled into a static Rust binary.

sigmoid when a probability-like score is needed for later combination. The result is static, easy to inspect, and fast enough for always-on use. There is no model server, no ONNX dependency, and no promptable reasoning loop inside the screening layer. In the measured harness, the L1 path runs in sub-millisecond time, while the Prompt Guard 2 baseline operates at a median of 49 ms and a p95 of 324 ms on the same machine.

The choice of character n-grams rather than word or subword features is deliberate. A character n-gram window slides over the raw string ignoring linguistic boundaries, which means it sees through common evasion tactics that word tokenization cannot: spaced-out characters (`s u d o`), Base64 fragments (`eyJ`, `==`), hex encoding (`%20`, `\x0a`), and Unicode substitution all leave contiguous character-level traces even when word boundaries are destroyed. The tradeoff is vocabulary size—an unconstrained character n-gram space over a large corpus can reach tens of millions of features—which is why the production configuration caps features at 15,000 and requires `min_df=5`. Early specialist experiments confirmed the choice empirically: word n-grams captured roleplay and instruction-override patterns well but missed the structural attack families (adversarial suffixes, obfuscation, indirect injection) that character n-grams resolved.

This design was further informed by those specialist experiments. We initially explored ensembles of category-specific models, each tuned for a different attack family (e.g., word n-grams for roleplay jailbreak, character n-grams without word boundaries for adversarial suffix). Those experiments showed that the attack categories were real enough to support specialized signal, but they also revealed the cost of runtime specialization. Each additional specialist increased curation burden, wiring complexity, and latency. That pushed the architecture back toward a single strong generalist for L1. The specialist work did not disappear, however. Its attack decomposition became the eight-category reason taxonomy later used by Mirror: instruction override, roleplay jailbreak, meta-probe, exfiltration, adversarial suffix, indirect injection, obfuscation, and constraint bypass.

Mirror is a system to arrange data that enables building a first-pass detector that fits a layered security system.

4 The Mirror Design Pattern

This section isolates the geometry argument. Its role is to show, through the `v2`→`v3` transition, that changing corpus structure while keeping the model family in place materially changed the operating point.

4.1 Why Geometry Matters

A linear classifier learns a boundary between positive and negative examples. If the training distribution lets it separate those classes using language, formatting, topic, or source artifacts, it will do that instead of learning the structure of prompt injection. This is normal empirical risk minimization on a badly shaped dataset.

Mirror architecture organizes the training corpus into cells of matched positive and negative examples, forcing a linear classifier to learn the boundary between them. In practice, that means aligning malicious and benign examples across dimensions such as language, format, length, and reason, so that those dimensions stop serving as spurious separators. Mirror is not just balancing class counts. It is an attempt to control the geometry of the training space.

This became the central hypothesis of the project after two earlier observations. The first was leakage. Early results were inflated because the data pipeline was not yet provenance-strict enough, and train–evaluation contamination boosted performance. The second was representational skepticism: if dense embedding spaces did not yield a clean linear boundary, perhaps the obstacle was the representation and corpus geometry rather than the task itself.

4.2 Mirror Construction

Mirror uses explicit cells as the unit of curation. A Mirror cell is a curated subset of the training corpus indexed by attack reason and language, containing matched positive and negative examples subject to explicit admissibility and provenance rules. Matching aligns nuisance dimensions—format, topic, approximate length—so that the remaining discriminative signal is the control-plane attack pattern rather than a corpus artifact. Examples that cannot honestly satisfy a cell's contract are routed to residual or background pools rather than counted as occupied.

In the `v5` checkpoint, cells are defined over eight attack reasons and four languages for a 32-cell target topology. Each source is annotated with scope and provenance rules.

This organization required software, not only labeling. `parapet-data` provides the provenance pipeline and `parapet-runner` the reproducible experiment runner.[1] Together they track source hashes, semantic parity, manifests, and cell fills. That infrastructure is not incidental. It is part of the method. Without it, the project would have repeated the same leakage and false-occupancy mistakes that partially motivated Mirror in the first place.

4.3 v2 to v3: Geometry as the Active Variable

The clearest evidence for Mirror came from the `v2` to `v3` transition. With the same model family, the project moved from a weaker geometry to a much stricter source-locked Mirror layout. The historical `v2` holdout result was already competitive, reaching 0.835 F1 with 0.931 recall on a 2,386-case holdout and exceeding the transformer baselines on recall. But it still carried too many false positives and too much residual impurity in the curation logic.

The `v3` result was the real turning point. At the natural SVM margin threshold of $t = 0.0$, the cleaned source-locked mirror baseline reached 0.918 F1 (precision 0.932, recall 0.905, 62 FP, 90 FN on a 1,875-case holdout). After further cleaning, this improved to 0.926 F1 (precision 0.945, recall 0.908, 50 FP, 87 FN). The `v2` mirror result on the same model family had been 0.835 F1 with 356 false positives. The `v3` mirror cut false positives by more than 80% and gained 8–9 F1 points by reorganizing the same corpus into stricter source-locked cells. The important point is not one decimal. It is the direction of change under a fixed model family. Data geometry materially changed the operating point.

The mixed-ratio ablation from `v2` supports the same interpretation. These absolute F1 values reflect the `v2`-era corpus and holdout; the important signal is the monotonic relationship, not the

[1] https://github.com/Parapet-Tech/parapet

Table 2: Mixed-ratio ablation (`v2` corpus, three seeds). Performance degrades monotonically as Mirror fraction decreases.

Mirror : Non-mirror	Holdout F1 (mean ± std)
100:0	0.837 ± 0.005
70:30	0.825 ± 0.004
50:50	0.819 ± 0.008
30:70	0.819 ± 0.003
0:100	0.788 ± 0.002

scalar levels. When the training data was blended from pure Mirror (100:0) to pure non-mirror (0:100) across three seeds, holdout F1 degraded monotonically (Table 2).

No sweet spot emerged from blending. Pure Mirror stayed best across all seeds. This is exactly the pattern one would expect if geometry, not just volume, were carrying the gain.

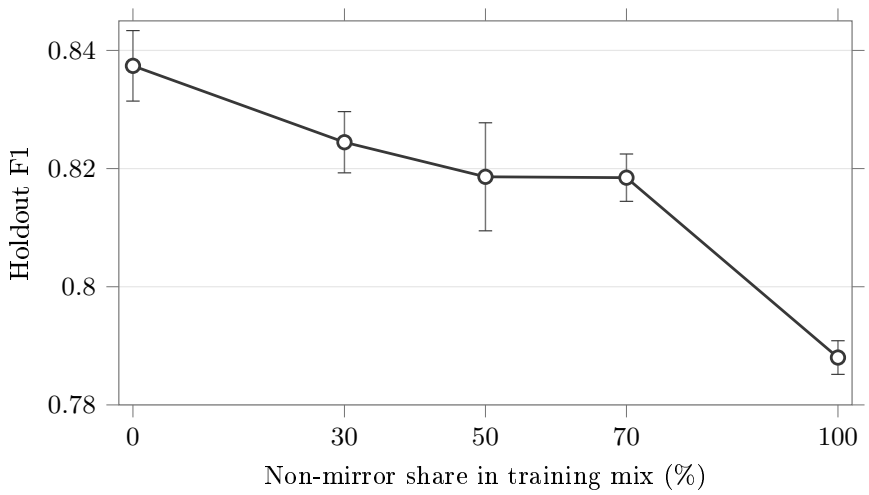

Figure 3: Holdout F1 as a function of Mirror-to-non-mirror training ratio (`v2` corpus, three seeds). Pure Mirror (100:0) is consistently best; performance degrades monotonically.

4.4 What the Model Learns

The feature story matters because it explains why Mirror is useful rather than magical. Weakly structured baselines tended to learn keyword fragments and substrings of "secret," "password," and "confidential." Those features detect prompts that talk about sensitive things, which is a proxy for the task but not the task itself. Mirror-trained models were pushed toward injection mechanics instead: instruction override cues, role and control-plane markers, adversarial delimiters, and the structural traces of attempted hijacking. This is still sparse lexical modeling, not semantics. But it is far closer to the real task than a model that memorizes the vocabulary of confidentiality.

The control-plane versus data-plane distinction is part of the method, not an incidental cleanup pass. Mirror aggressively strips subjective content-safety violations from the attack corpus so that L1 is trained to police control-plane attacks—instruction hijacking and authority transfer—rather than downstream data-plane categories such as toxicity, copyright, or other policy misuse.

That design choice is one reason the resulting decision boundary is cleaner. A classifier that is not asked to double as a general content-safety filter can spend its capacity on the mechanics

of prompt takeover instead of on policy-adjacent lexical cues. In our architecture, content safety sits outside the intended L1 scope: it is broader, more subjective, and better handled by later semantic layers than by the first-pass classifier.

The contamination story belongs here as well. Leakage was not a side note; it changed the design of the project. The provenance pipeline exposed 18.6% train–evaluation contamination in the non-mirror baseline and 7% in the early mirror corpus. Without content hashing and manifest tracking, the baseline would have appeared to crush the evaluation because it had memorized nearly a fifth of the test set. The fix was to carve the evaluation holdout first and assert zero train–eval intersection via content hash before every run. Once those leaks were repaired and the experiment rerun on clean splits, Mirror stopped being just a clever sampling trick and became a stricter empirical contract. That contract is what made later claims believable.

5 Validity Recovery and Multilingual Closure: v5

This checkpoint asks a different question from `v3`: not whether Mirror can improve results under a fixed model family, but whether the method still holds after much stricter validity, source-contract, and multilingual coverage requirements are imposed.

Between `v3` and `v5`, error-driven corpus review showed that content-safety contamination was a substantial component of the source-pool defects. Across 167 adjudicated drops in the cleanup ledger, 34.1% were explicit content-safety violations mislabeled as prompt injection; this rises to 35.9% when adjacent policy-not-PI categories such as copyright, PII, and academic dishonesty are included. In the stricter `v5_r1` review tranche alone, explicit content-safety cases accounted for 42.2% of adjudicated drops. This motivated a ground-up rebuild at lower volume with stricter label contracts.

That finding changed how we interpreted the earlier result. `v3` had shown that mirror geometry mattered, but it had not yet imposed the validity contract we ultimately wanted. For `v5`, we scaled down to curate a smaller corpus we could defend cell by cell. In practice, 5,000 was not chosen because performance had clearly plateaued there; it was the largest public-source corpus we could support while still filling 31 of 32 cells under the stricter validity contract. By this point it had become clear that occupancy was not enough. A cell with one mislabeled row is not coverage. A multilingual pooled source is not automatically valid mirror evidence for every reason–language pair it touches. If the project was going to claim strict data geometry, it had to stop hiding behind loose notions of coverage.

The `v5` recovery work formalized this. Coverage was redefined as quota-based occupancy under explicit source contracts. The active geometry was 8 reasons $\times$ 4 languages $=$ 32 logical cells. Each cell needed reason-grounded malicious coverage and valid benign support. Rows from pooled or mixed-scope sources could still be useful, but they had to be routed as residual, background, or quarantine unless they were promoted through a documented review path.

This made the data problem harder, not easier. The first `v5` run covered only 18 of 32 cells; the final recovery line closed 31 of 32. The recovery logic was staged rather than monolithic: broad passes were used first to recover dense lanes and expose real deficits, then tiny reviewed micro-sources and per-source tail extraction were used to close sparse cells one by one. Chinese and Arabic were recovered through that combination of broad sweeps and micro-sources. Russian proved harder and shifted to tail-queue review across 444 rows from 10 source-specific queues, which closed `indirect_injection`, `exfiltration`, and `meta_probe` but still left `RU constraint_bypass` as the single accepted miss after direct review found only 2 marginal candidates across the available sources (0.5% hit rate).

The Mirror method still worked after the evaluation contract was tightened, the sources audited, and the gaps named rather than buried.

The final matrix is also useful as a compact demonstration of what Mirror means in practice.

Reason/language	EN	RU	ZH	AR
instruction override	OK 234/199	OK 31/27	OK 25/21	OK 22/19
roleplay jailbreak	OK 234/199	OK 31/27	OK 25/21	OK 22/19
meta probe	OK 234/199	OK 31/27	OK 25/21	OK 22/19
exfiltration	OK 234/199	OK 31/27	OK 25/21	OK 22/19
adversarial suffix	OK 234/199	OK 31/27	OK 25/21	OK 22/19
indirect injection	OK 234/199	OK 31/27	OK 25/21	OK 22/19
obfuscation	OK 234/199	OK 31/27	OK 25/21	OK 22/19
constraint bypass	OK 246/199	M 0/27	OK 25/21	OK 41/19

Counts are mal/ben. **M** marks the accepted miss.

Figure 4: Final `v5` coverage matrix: 8 reasons × 4 languages = 32 cells. Green = closed with valid data; red = `RU constraint_bypass`, the single accepted miss.

It is not just abstract geometry. It is a concrete multilingual corpus with explicit reason slots, explicit failures, and explicit provenance. That is a stronger scientific object than a single scalar F1.

6 Experimental Results

The evaluation contract for the final 5k comparison is summarized in Table 3.

6.1 Main 5k Result

The current 5k control result comes from the final 5,000-sample corpus built in the `v5` recovery line and a hyperparameter sweep over the linear SVM family. The best reported run uses character word-boundary n-grams with n from 3 through 5, `max_features=15000`, `min_df=5`, and $C = 1.0$, and reaches 0.9207 F1, 0.8848 precision, and 0.9597 recall on a 524-case held-out evaluation set with 248 malicious and 276 benign examples. At threshold 0.0 this corresponds to 31 false positives and 10 false negatives. Bootstrap 95% confidence intervals (10,000 resamples) are $[0.895, 0.943]$ for F1, $[0.844, 0.921]$ for precision, and $[0.934, 0.983]$ for recall. This is the clearest final demonstration of the paper's empirical point: a small, static, carefully curated linear model can become very strong when the data geometry is right.

The cryptographic content hashing and provenance trail matter here because good numbers without provenance are not enough.

6.2 Regex-Only Baseline

The natural first question is whether machine learning is needed at all. Our L3 regex layer—75 hand-written patterns covering 10 attack categories—reaches 99.2% precision but only 14.1% recall (F1 24.7%) on the same evaluation corpus. Regex is excellent at catching known attack

Table 3: Evaluation protocol for the `v5` 5k comparison.

Protocol item	Setting
Holdout artifact	524 cases total: 248 malicious, 276 benign
Split discipline	Holdout carved and frozen before training; training and evaluation are enforced content-hash disjoint
Evaluation preprocessing	L0 sanitization enabled at evaluation time for all compared detectors
Holdout composition	The 524-case holdout spans the same four-language, eight-reason task space as the training corpus, though not with uniform occupancy
Confidence intervals	95% bootstrap intervals with 10,000 resamples over the 524-case holdout
Threshold selection	L1 main results at fixed t=0.0 (natural SVM margin cutoff); PG2 evaluated at default thresholds (`pg_threshold=0.5`, `block_threshold=0.5`)
"Open-source" in release terms	Training and evaluation examples are drawn from public-source corpora; the release publishes code, specs, manifests, and hashes needed to reconstruct the benchmark, not redistributed raw corpora or local runner artifacts

templates with near-zero false positives, but prompt injection is too varied for static rules. The gap between 14% recall and 96% recall is exactly what the learned L1 boundary provides.

6.3 Semantic Baseline

The baseline protocol was designed to keep the comparison operationally fair and mechanically simple. We evaluate `pg2-22m` off the shelf, without task-specific fine-tuning, through the same `parapet-eval` harness on the exact same 524-case holdout artifact used for the SVM results, remapping the L2a decision to the binary screening label. Both paths run with L0 sanitization enabled at evaluation time. For PG2 we use the standard `pg_threshold=0.5` and `block_threshold=0.5` configuration; for L1 we report the compiled classifier at threshold t=0.0, the natural margin cutoff used in the main holdout tables. We choose the smaller `pg2-22m` variant because the comparison target is hot-path screening, where latency is part of the result; we do not present it as an upper bound on all possible semantic baselines. The earlier ProtectAI DeBERTa baseline is not carried into the main table because it was only evaluated in the historical `v2` protocol on a different 2,386-case holdout, where it underperformed materially (0.539 F1, 0.615 precision, 0.480 recall). After that result, direct-comparison effort focused on Prompt Guard 2 as the stronger semantic comparator. We also evaluated the 86M-parameter PG2 variant; its recall profile was nearly identical to the 22M model at four times the parameter count, confirming the smaller variant as the relevant comparator.

6.4 Head-to-Head Comparison

On the same 524-case holdout, the compiled L1 path ran at 0.32 ms mean latency (0.13 ms median, 1.40 ms p95) in the eval harness, with earlier head-to-head runs landing around 0.26 ms mean. Meanwhile the `pg2-22m` Prompt Guard 2 baseline reached 0.5914 F1, 0.8871 precision, and 0.4435 recall, with 14 false positives and 138 false negatives. The PG2 bootstrap intervals are $[0.529, 0.648]$ for F1, $[0.828, 0.941]$ for precision, and $[0.381, 0.506]$ for recall. The recall and F1 intervals are well separated, while the precision intervals overlap. PG2 is conservative and precise on this set, but it misses most malicious cases. The Mirror-trained L1 model gives up some precision in exchange for a very large recall gain. For a first-pass detector in a layered

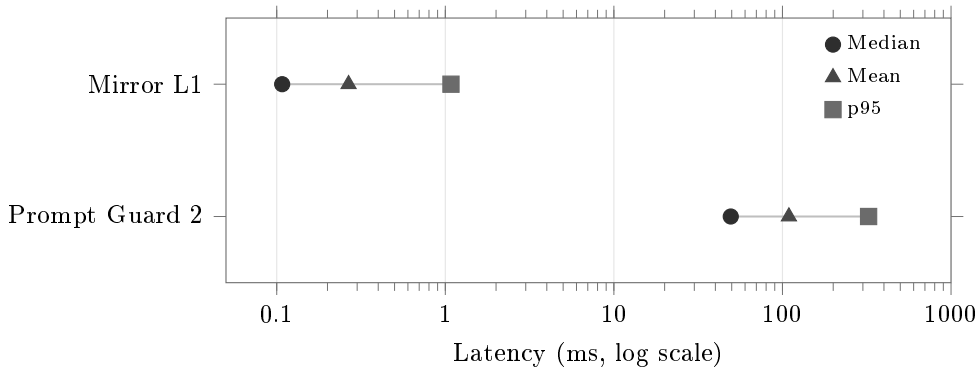

Figure 5: Latency distributions for L1 (compiled SVM) vs. PG2 (22M). L1 operates in sub-millisecond time; PG2 has a long tail reaching 324 ms at p95.

system, that trade is often correct.

The three approaches occupy different points on the same trade-off surface (Table 4).

Table 4: Head-to-head comparison on the 524-case `v5` holdout.

Detector	Prec.	Recall	F1	FP	FN	Latency
L3 regex (75 patterns)	0.992	0.141	0.247	—	—	<1 ms
Mirror L1 SVM (5k, t=0.0)	0.885	0.960	0.921	31	10	<1 ms
Prompt Guard 2 (22M)	0.887	0.444	0.591	14	138	49 ms

This comparison should be framed carefully. Once L1 has been rebuilt around strict geometry, the semantic baseline no longer looks like the obvious best answer for screening. It becomes a slower, narrower, and more expensive signal. And the regex layer, while essential for catching known templates at near-zero cost, cannot replace a learned boundary. That is exactly why the open question is now about replacing or reducing L2a rather than depending on it by default.

6.5 Latency and Operational Profile

The latency story is part of the result, not just an implementation detail. In the measured harness, the L1 path operates in sub-millisecond time, with historical measurements on the order of a few tenths of a millisecond end to end. The PG2 baseline on the 5k holdout had a mean latency of 109.2 ms, a median latency of 49.4 ms, and a p95 latency of 324.4 ms. Those numbers imply a different deployment model.

The same is true of the dependency profile. L1 is compiled into the Rust binary as static weights. PG2 requires a model runtime and carries a larger, more fragile systems footprint. Even if the raw classification numbers were closer, this difference would still matter. Security layers are operational systems, not just classifiers.

6.6 Reproducibility and Release

The reproducibility story is part of the contribution. The curation pipeline records source hashes, cell fills, gap reports, and semantic hashes through `parapet-data`. The experiment runner records training configuration, runtime hashes, evaluation output, and semantic parity through `parapet-runner`. In practical terms, the paper is supported by a toolchain that can rebuild named corpora from versioned specs and rerun named experiments from recorded manifests and configs rather than relying on hand-maintained notebooks or ad hoc scripts. Compact specs

expand into full cell-level specifications, making the resulting mirror layout auditable rather than implicit.

The curation pipeline (`parapet-data`) and experiment runner (`parapet-runner`) are released under the Apache 2.0 license at https://github.com/Parapet-Tech/parapet. The `v5` results reported in this paper correspond to the `mirror-paper-v5` tag. Because training and evaluation data are drawn from third-party public-source corpora with varying redistribution terms, the release includes specs, manifests, and content hashes sufficient to reconstruct the exact corpus and holdout from their original sources, rather than redistributing the raw data directly.

6.7 Challenge-Set Honesty

The model does have a ceiling. On an adversarial hard-benign challenge set of 2,386 security-adjacent documents—whitepapers, CTF writeups, red-team documentation that quotes or discusses attack patterns—the L1 SVM false-positive rate at threshold 0.0 was 51.9%. Even at a raised threshold of +2.0, the hard-benign FPR remained 12.3%. On the same challenge set, PG2 (22M) reached 21.3% FPR—better, but still far too high for production use. A 22-million-parameter semantic model cuts the hard-benign FPR roughly in half relative to a sparse lexical model, but it does not solve use-versus-mention disambiguation. A separate challenge slice drawn from JailbreakBench [9] paraphrase variants initially appeared to expose a shared ceiling (∼21% recall for both the SVM and Prompt Guard 2). Post-hoc inspection, however, showed that this slice consists of direct harmful-content requests rather than prompt injection: bomb-making, identity theft, hate speech, and similar content-safety refusals, with no instruction override, role hijacking, or indirect injection. We therefore no longer interpret that result as a PI benchmark. Its only useful lesson is methodological: evaluation slices must be scoped strictly to control-plane manipulation.

This is also the right bridge to the next design problem. The question is no longer whether to replace the SVM with a bigger model. It is how much of the current L2a residual can be replaced by a non-promptable or at least non-generative second pass.

7 Limitations

This paper is a proof of architecture and curation method, not a claim that 5,000 samples are the final production recipe for prompt injection detection. The project hit a real open-source data ceiling, especially in multilingual attack coverage. The final `v5` matrix closed 31 of 32 cells, but it did so by careful sourcing and micro-source construction, and one Russian lane remained an explicit accepted miss. We therefore do not claim that performance plateaued at 5,000. It was the largest corpus we could gather under the final public-data validity contract. That is a success for methodological honesty, not evidence that the data problem is solved.

The feature family also has clear limits. Character n-grams are very good at capturing lexical and structural control-plane traces, but they do not robustly resolve contextual ambiguity. A security whitepaper quoting a jailbreak prompt can still look too much like an attack (51.9% FPR on the hard-benign challenge set at default threshold). An earlier evaluation against JailbreakBench paraphrase variants appeared to show both models plateauing at ∼21% recall, but that slice turned out to be content-safety material rather than prompt injection (see Section 6.4). True paraphrase-robust PI evasion—attacks that preserve injection semantics while eliminating lexical n-gram traces—remains an open and likely architectural challenge for any sparse linear model.

Although we report bootstrap intervals, the final holdout is modest in size, and broader external validity across unseen production traffic remains unproven.

The evaluation scope is also bounded by the datasets we were able to build and audit. Public prompt injection datasets are noisy. Some mix content safety with prompt injection. Some have

poor provenance. Some overrepresent English or a few familiar attack families. Mirror mitigates this by making source contracts explicit, but it does not magically produce a gold-standard universal benchmark.

Finally, the paper does not resolve the whole layered architecture. It gives a stronger answer for L1 than we had before. It does not yet show how to retire the current L2a semantic model without losing too much on the residual set.

8 Conclusion

The main result of this paper is not that linear models beat transformers in the abstract. It is that for L1 prompt injection screening, strict data geometry can matter more than model scale. Once the training corpus is organized as matched positive and negative cells, with provenance and validity rules enforced as part of the method, a simple sparse linear classifier becomes far more capable than the standard story would predict.

The `v2` checkpoint established that this classifier might belong inside a layered defense architecture. The `v3` checkpoint showed that the geometry itself moved the decision boundary. The `v5` checkpoint showed that the method could survive validity pressure, multilingual recovery, and explicit source-contract accounting. The final 5k result, 0.9207 F1 and 0.9597 recall on held-out data, is therefore best understood as evidence for a design pattern.

The remaining problem is now clearer than it was at the start. Within the scope of this study, the first layer is now empirically supported. The open question is what should replace or reduce the current L2a semantic layer. The honest answer is probably not "nothing." L1 still leaves a residual set defined by contextual ambiguity, use-versus-mention cases, and semantically thin attacks. The next research target is therefore a better residual architecture: one that preserves the operational and security advantages of Mirror at L1 while shrinking the need for a slow, promptable semantic detector at L2a.

If that problem can be solved, the generalization claim becomes much stronger. But even without it, these results already suggest that improving hot-path prompt injection defense may depend less on larger hot-path models and more on better architecture, stricter geometry, and semantic reasoning reserved for the cases that truly require it.

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